Task-10:- Technical Documentation(Create a Custom Dashboard in CloudWatch)

1. Introduction

1.1 Purpose

The purpose of this technical document is to describe the implementation of a **CloudWatch monitoring dashboard for multiple EC2 instances** within the AWS environment of ABC Technologies.

This document explains the monitoring architecture, tools used, infrastructure design, and the implementation steps followed to create the dashboard.

The intended audience for this document includes:

- CTO
- DevOps Engineers
- Developers
- Solution Architects
- Cloud Operations Team

The document provides an overview of how infrastructure monitoring was implemented using AWS CloudWatch to provide **centralized visibility into system performance and health metrics**.

1.2 Scope

The scope of this document covers the **implementation of a centralized CloudWatch dashboard for EC2 infrastructure monitoring**.

The monitoring solution focuses on the following metrics:

- CPU Utilization
- Memory Usage
- Disk Usage
- Network In
- Network Out

The document explains the monitoring design, implementation process, and visualization techniques used to provide a **clear and business-friendly infrastructure monitoring interface**.

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2. Overview of DevOps Project

2.1 Definition

A DevOps monitoring solution integrates infrastructure monitoring, operational visibility, and performance tracking into a centralized platform.

In this project, **AWS CloudWatch** was used to monitor EC2 instances and visualize system metrics through a custom dashboard.

The key benefits of this monitoring solution include:

- Centralized infrastructure monitoring
- Improved operational visibility
- Faster identification of performance issues
- Better infrastructure management

2.2 Key Principles

The monitoring solution aligns with the **AWS Well-Architected Framework**, which focuses on six pillars:

Operational Excellence

Continuous monitoring of system metrics enables efficient operations management.

Security

Access to monitoring dashboards is controlled through AWS IAM roles and permissions.

Reliability

Infrastructure metrics allow quick detection of failures or abnormal behavior.

Performance Efficiency

System performance metrics help analyze resource utilization and optimize workloads.

Cost Optimization

Monitoring infrastructure helps detect inefficient resource usage.

Sustainability

Efficient resource monitoring helps reduce unnecessary infrastructure consumption.

2.3 Goals and Objectives

The main goals of this project are:

- Provide centralized monitoring of EC2 infrastructure.
- Allow easy comparison of performance across multiple instances.
- Improve visibility into system health.
- Enable quick identification of unusual infrastructure behavior.

2.4 Constraints and Assumptions

Constraints

Time Constraint
The task had a completion time of **3 hours**.

Technical Constraints
Memory and disk usage metrics require installation of the **CloudWatch Agent**.

Visualization Constraints
Some visualization options available in tools like Grafana are limited in CloudWatch.

Assumptions

- All EC2 instances are running within the same AWS account.
- Monitoring permissions are configured correctly.
- CloudWatch Agent is installed where required.

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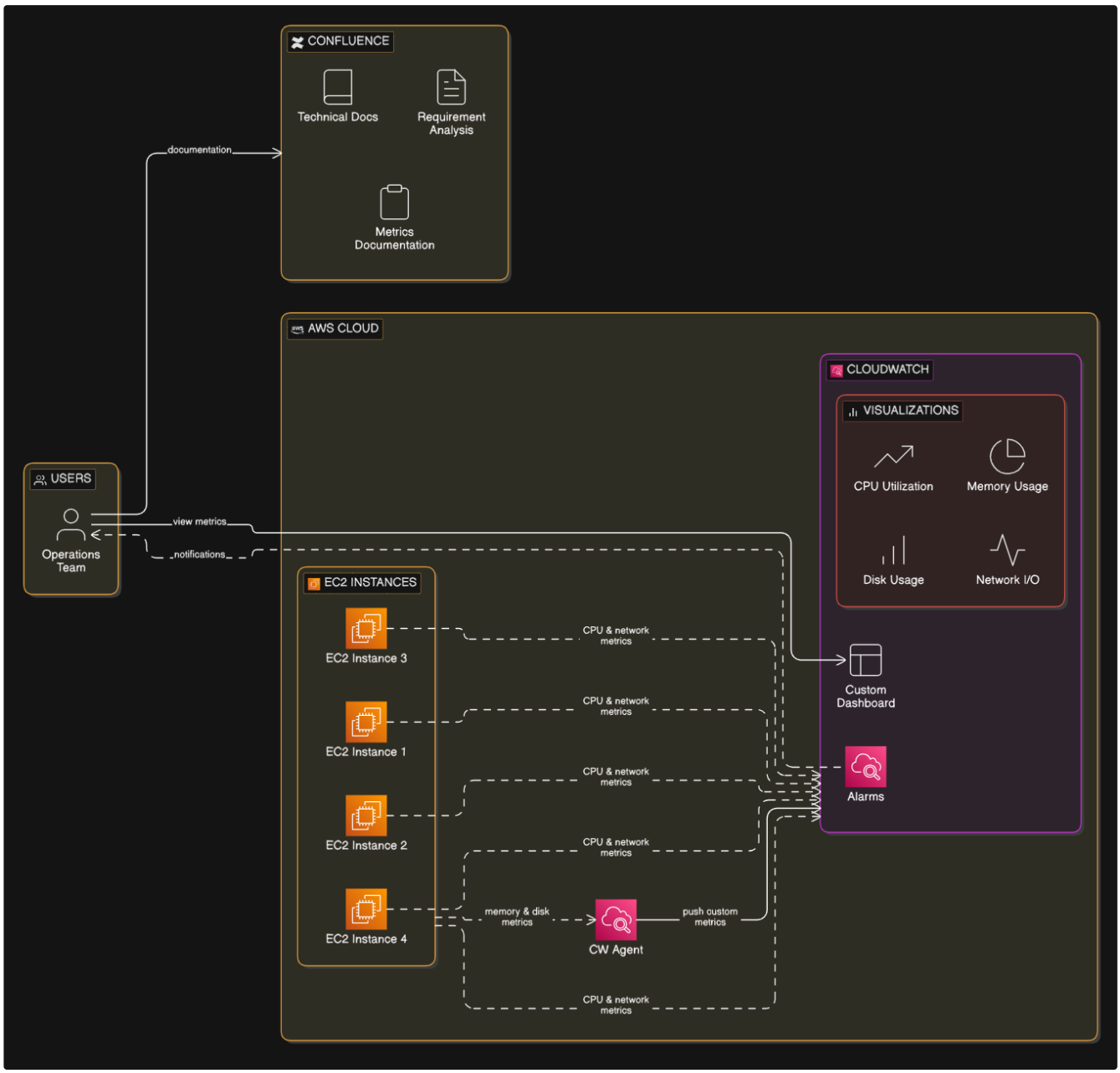
3. Architecture and Infrastructure



3.1 Infrastructure Overview

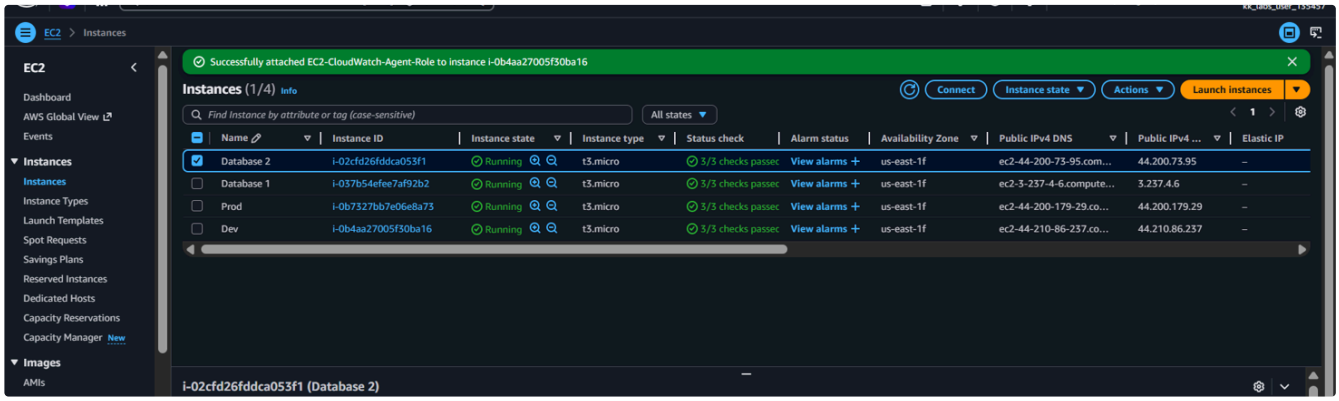
The infrastructure monitoring system consists of the following components:

Component	Description
AWS EC2	Virtual machines running application workloads
CloudWatch	Monitoring service collecting infrastructure metrics

CloudWatch Dashboard	Visualization platform for metrics
CloudWatch Agent	Collects memory and disk metrics

The CloudWatch dashboard collects metrics from **four EC2 instances** and displays them through different visualization widgets.

EC2 Instances Running:-



3.2 Environments

The monitoring dashboard supports environments that include:

Development Environment
Used for testing application changes.

Testing Environment
Used to verify system performance and application stability.

Production Environment
Used for running live application workloads.

Monitoring ensures all environments remain **stable and operational**.

3.4 Application Architecture & User Request Flow

The monitoring architecture works as follows:

1. EC2 instances generate infrastructure metrics.
2. CloudWatch collects and stores these metrics.
3. CloudWatch Agent gathers system-level metrics such as memory usage.

4. CloudWatch Dashboard visualizes metrics through different widgets.

CloudWatch Dashboard:-

3.3 Tools and Technologies

The following tools were used in the monitoring implementation:

Tool	Purpose
AWS EC2	Host application workloads
AWS CloudWatch	Infrastructure monitoring
CloudWatch Dashboard	Visualization interface
CloudWatch Agent	Collect memory and disk metrics

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4. Development Practices

4.1 Version Control

Infrastructure monitoring configuration and documentation are maintained using version-controlled repositories to track changes and maintain consistency.

4.2 Continuous Integration

Monitoring solutions support DevOps workflows by providing continuous infrastructure performance monitoring, enabling teams to identify issues during development and testing.

4.3 Automated Testing

Infrastructure monitoring helps validate system stability by continuously observing resource utilization metrics.

4.4 Code Quality and Analysis

Monitoring tools help detect performance bottlenecks and ensure system stability during deployment cycles.

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5. Deployment and Release Management

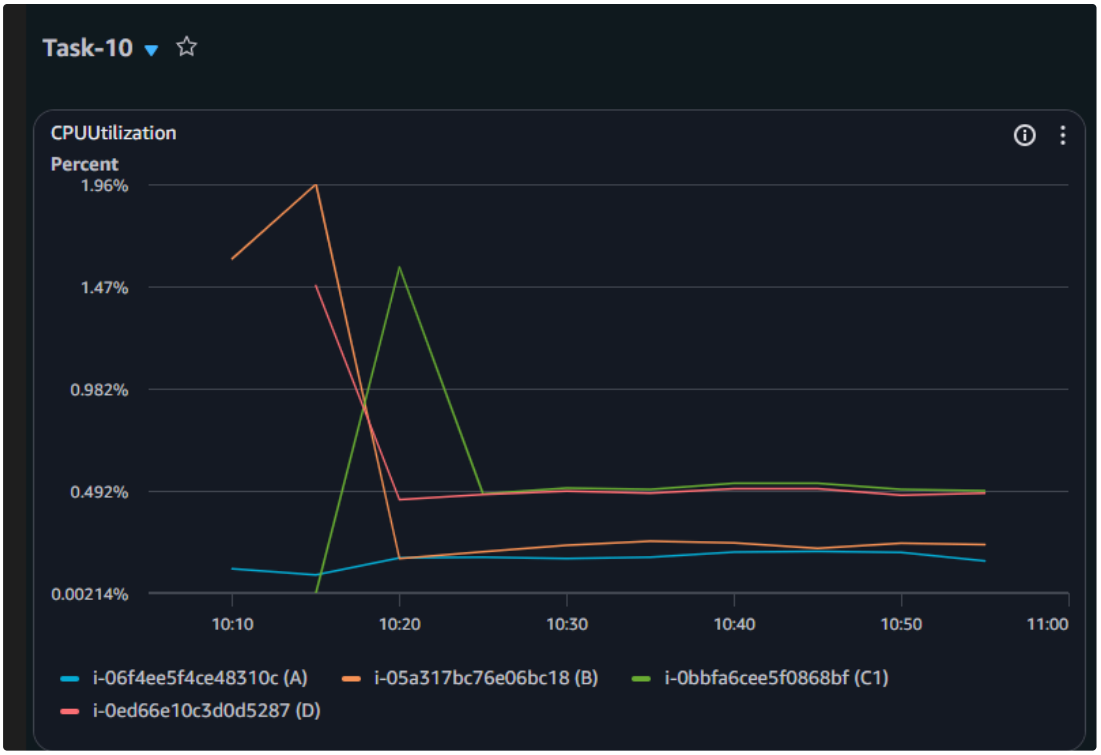
5.1 Deployment Strategy

The monitoring dashboard was deployed directly through the **AWS CloudWatch Dashboard interface**.

The deployment steps included:

1. Accessing AWS CloudWatch.
2. Creating a new dashboard.
3. Adding infrastructure monitoring widgets.
4. Configuring metrics and visualizations.

CPU Utilization Widget



5.2 Configuration Management

Infrastructure configuration is managed through AWS services, ensuring monitoring settings remain consistent across instances.

5.3 Continuous Delivery / Deployment

Monitoring dashboards can be updated or modified dynamically without affecting running infrastructure services.

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6. Monitoring and Incident Management

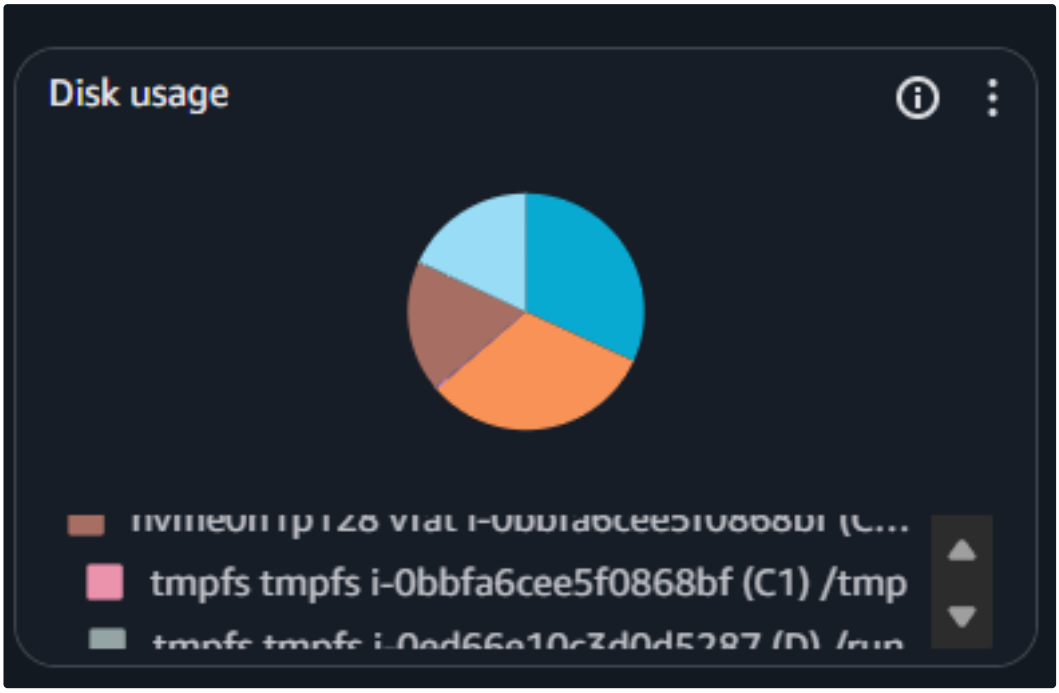
6.1 Monitoring Tools and Metrics

The monitoring solution uses **AWS CloudWatch** to track key infrastructure metrics.

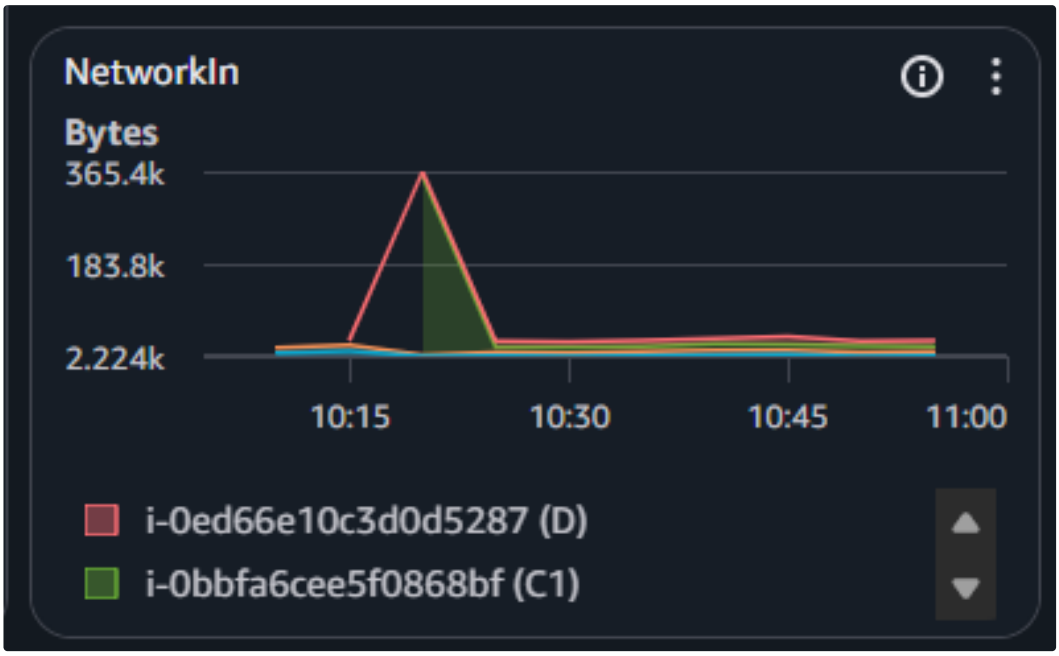
Monitored metrics include:

- CPU Utilization
- Memory Utilization
- Disk Usage
- Network In
- Network Out

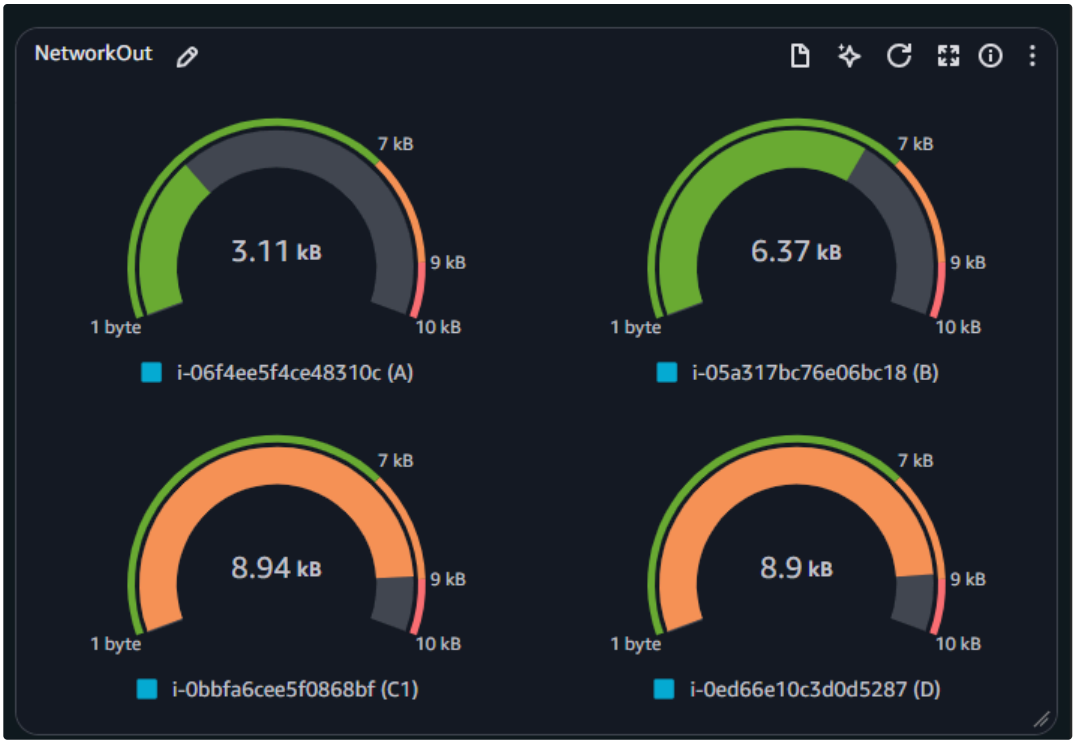
Disk Usage Widget



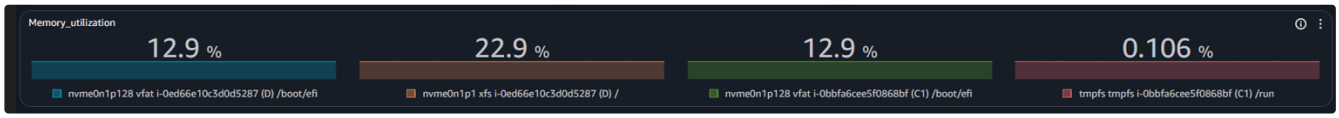
Network In Widget:-



Network Out Gauge



Memory Utilization Metrics



6.2 Alerting and Notification

CloudWatch can be integrated with **CloudWatch Alarms** to notify teams when specific thresholds are exceeded.

Notifications can be sent through:

- Email
- SNS
- Incident management tools

6.3 Incident Management

In case of abnormal infrastructure behavior:

1. CloudWatch metrics identify the issue.
2. Operations teams investigate the instance.
3. Corrective actions are applied.

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7. Security & Compliance

Security practices implemented include:

- IAM role-based access control
- Restricted access to monitoring dashboards
- Secure AWS account authentication
- Infrastructure monitoring to detect unusual activity

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8. Cost Optimisation

Monitoring infrastructure usage helps organizations optimize costs.

Benefits include:

- Identifying underutilized instances
- Detecting unnecessary resource consumption
- Improving infrastructure efficiency

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10. Implementation Timeline

10.1 Milestones and Deliverables

Phase 1 – Requirement Analysis
Understanding monitoring requirements and identifying key metrics.

Phase 2 – Dashboard Design
Designing the layout and selecting visualization types.

Phase 3 – Implementation
Creating CloudWatch widgets and configuring metrics.

Final Delivery – CloudWatch Monitoring Dashboard.

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11. Continuous Improvement

11.1 Feedback and Retrospectives

Feedback from operations teams can help improve dashboard usability and visualization clarity.

11.2 Iterative Enhancements

Future improvements may include:

- CloudWatch Alarms
- Advanced monitoring dashboards
- Automated incident notifications

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12. Conclusion

12.1 Summary

The CloudWatch monitoring dashboard provides a centralized solution for monitoring EC2 infrastructure.

By visualizing system metrics such as CPU usage, memory consumption, disk usage, and network traffic, the dashboard allows operations teams to quickly assess system health and respond to potential issues.

12.2 Next Steps

Future enhancements may include:

- Automated alerting systems
- Integration with DevOps monitoring pipelines
- Infrastructure performance analytics

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13. Customer Acknowledgement

Name: _____

Title: _____

Signature: _____

Date: _____